CONVERSION OF NUMBER SYSTEM

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CONVERSION OF NUMBER SYSTEM

REFERENCE BOOKS

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- Digital Design, M. Morris Mano, Michael D. Ciletti, Pearson Education, Inc.
- Digital Circuits and Design, S. Salivahanan, S. Arivazhagan, Oxford University Press.
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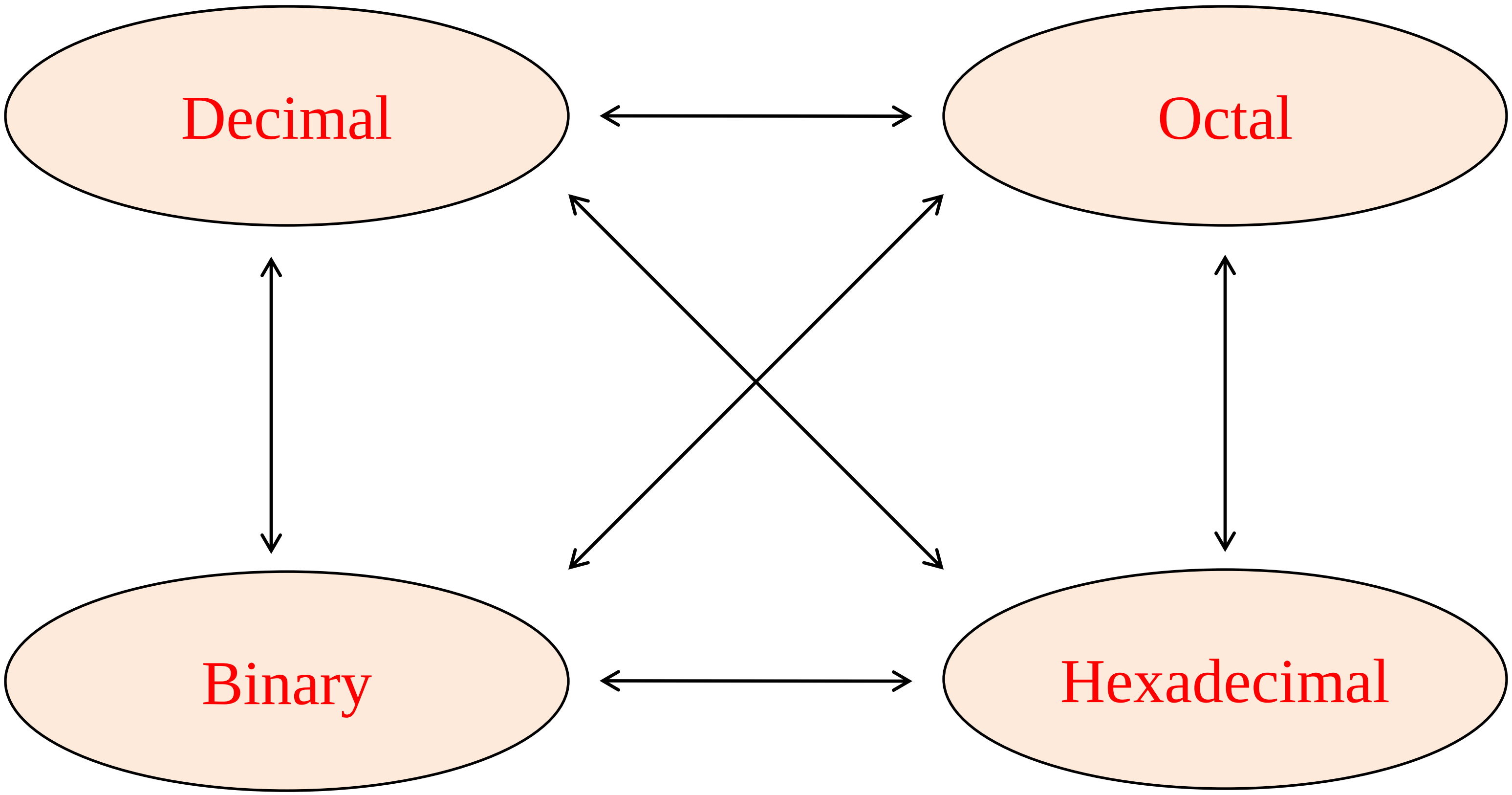
- Digital Electronics, Bignell James, Logic and Systems, Cengage Learning
- Digital Logic and Computer Design, M. Morris Mano, Pearson Education, Inc.

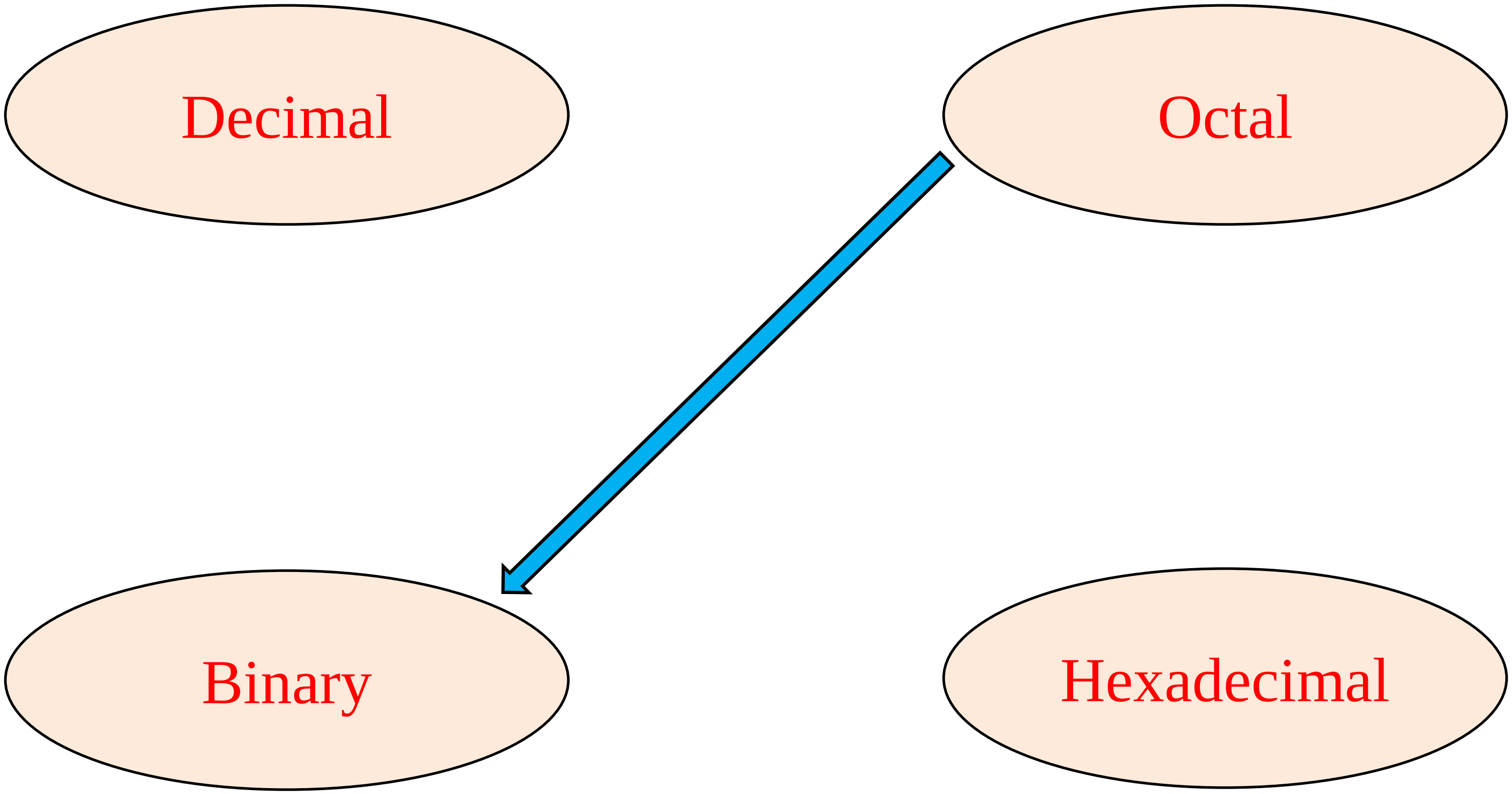
CONVERSION OF NUMBER SYSTEM

The possible conversions are



CONVERSION OF NUMBER SYSTEM

OCTAL TO BINARY CONVERSION:



CONVERSION OF NUMBER SYSTEM

PROCEDURE:

STEP1: Convert each octal digit to a 3 digit binary number (the octal digits may be treated as decimal numbers for this conversion).

STEP2: Combine all the resulting binary digits into a single binary number.

This method also known as 421 method.

Octal digit	3-bit Binary equivalent
0	000
1	001
2	010
3	011
4	100
5	101
6	110
7	111

CONVERSION OF NUMBER SYSTEM

EXAMPLE: Convert $(562)_8 = (?)_2$

Step 1: Convert each Octal digit to 3 binary digits

$5_8 = 101_2$

$6_8 = 110_2$

$2_8 = 010_2$

Step 2: Combine the binary group

$$562_8 = \underbrace{101}_{5}\ \underbrace{110}_{6}\ \underbrace{010}_{2}$$

Hence, Convert $(562)_8 = (101110010)_2$

CONVERSION OF NUMBER SYSTEM

BINARY TO OCTAL CONVERSION:

The bits of binary numbers are grouped into the groups of three bits starting at the LSB (starting from the right). Then convert each group of three binary digits to one octal digit.

EXAMPLE: $(101110)_2 = (?)_8$ and $(110111)_2 = (?)_8$

$$\frac{101}{421} \quad \frac{110}{421}$$

$$5 \quad 6$$

$$(101110)_2 = (56)_8$$

$$\frac{110}{421} \quad \frac{111}{421}$$

$$6 \quad 7$$

Thus, $(110111)_2 = (67)_8$

CONVERSION OF NUMBER SYSTEM

$$\text{Convert}\left(100110.11001\right)_2 = (\,?\,)_8$$

Solution :

$$\frac{100}{421}\ \frac{110}{421}\ .\ \frac{110}{421}\ \frac{010}{421}$$

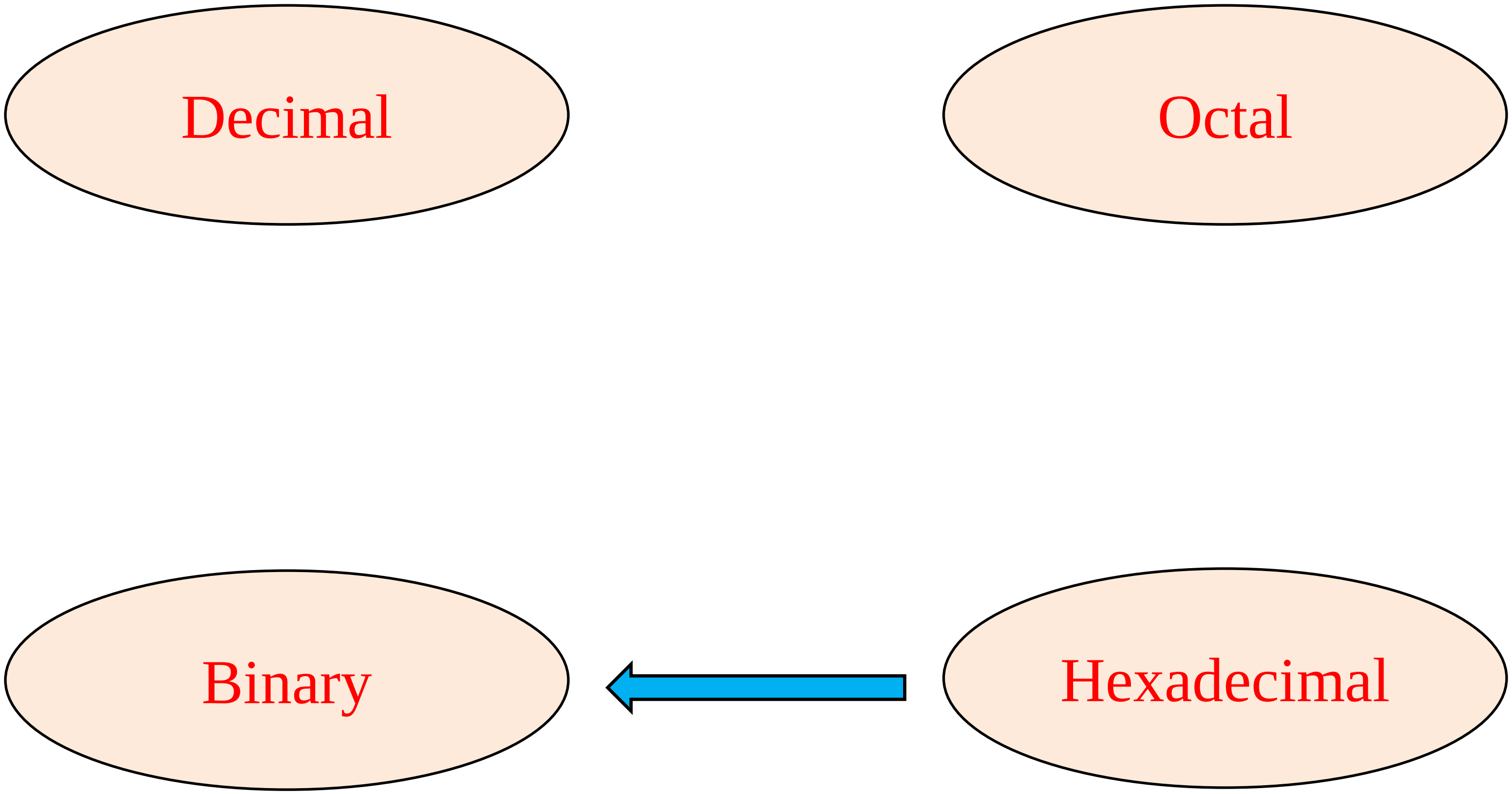
$$4\quad 6\qquad 6\quad 2$$

$$\left(100110.11001\right)_2 = (46\cdot 62)_8$$

Exercise: $(1111000.0001)_2 = (?)_8$ Ans: 170.04_8

CONVERSION OF NUMBER SYSTEM

HEXADECIMAL TO BINARY CONVERSION:



CONVERSION OF NUMBER SYSTEM

Step 1: Convert decimal equivalent of each hexadecimal digit to 4 binary digits.

Step 2: Combine all resulting binary groups (4 digits) into a single binary number.

This method is also known as 8421 method.

Decimal	Hexadecimal
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

CONVERSION OF NUMBER SYSTEM

Convert $(2AB)_{16} = (?)_2$ and $(ABC)_{16} = (?)_2$

Solution :

$2_{16} = 0010_2$

$A_{16} = 1010_2$

$B_{16} = 1011_2$

$$2AB_{16} = \frac{0010}{2}\ \frac{1010}{A}\ \frac{1011}{B}$$

Hence, $2AB_{16} = 0010101010\,11_2$

$$ABC_{16} = \frac{A\,(10)}{8421}\ \frac{B\,(11)}{8421}\ \frac{C\,(12)}{8421}$$

$$1010 \quad 1011 \quad 1100$$

Hence, $ABC_{16} = 1010101111\,00_2$

CONVERSION OF NUMBER SYSTEM

BINARY TO HEXADECIMAL CONVERSION:

The bits of binary number are grouped into group of 4 bits starting at the LSB. Then each group is converted to its hexadecimal equivalent.

EXAMPLE:

Convert $(11011110)_2$ into hexadecimal.

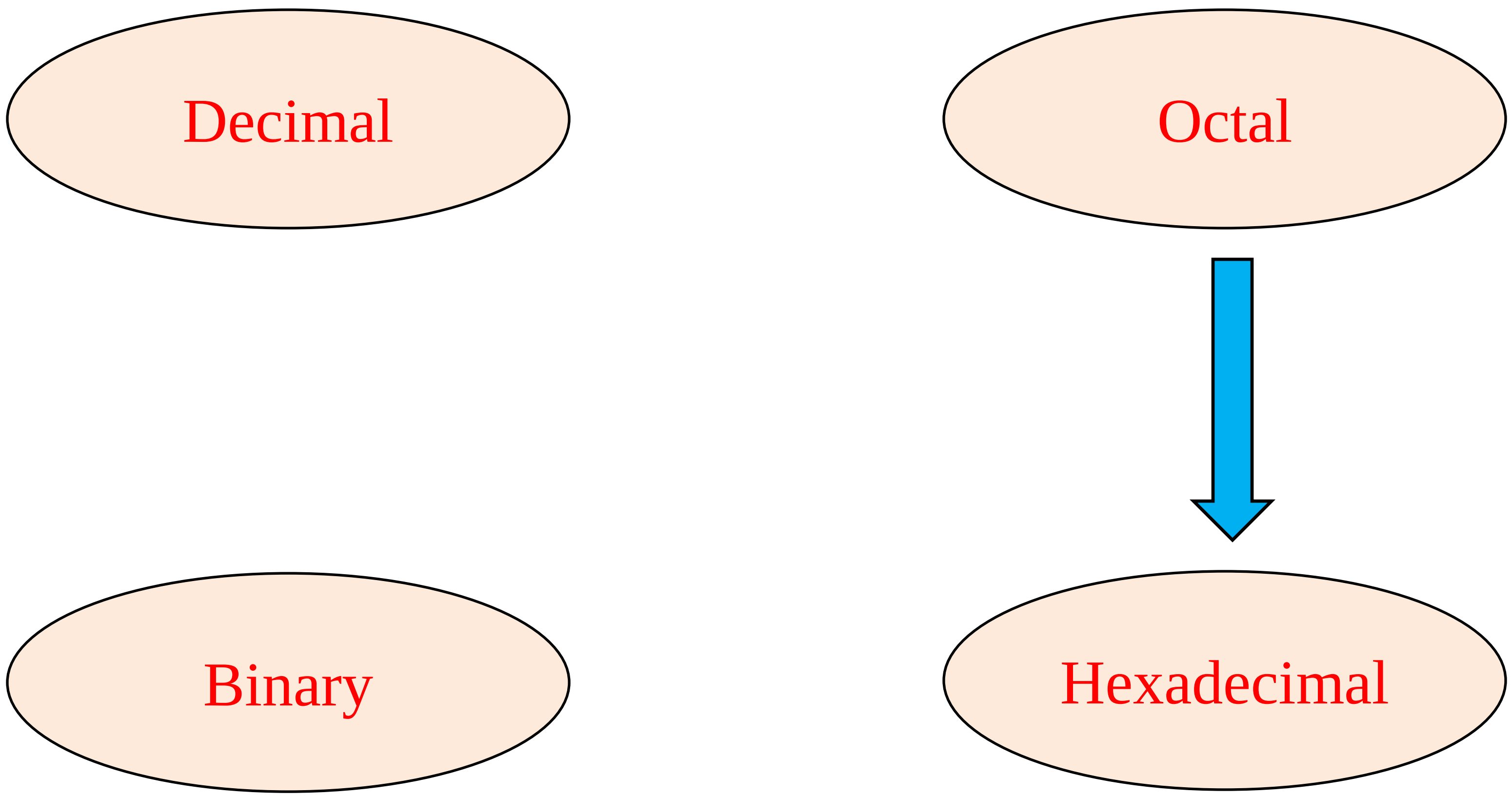
Solution :

$$\begin{array}{cc} 1101 & 1110 \\ \underline{8421} & \underline{8421} \\ D(13) & E(14) \end{array}$$

Thus, $(11011110)_2 = (DE)_{16}$

CONVERSION OF NUMBER SYSTEM

OCTAL TO HEXADECIMAL CONVERSION:



CONVERSION OF NUMBER SYSTEM

Step 1: Convert given Octal number into binary number

Step 2: The bits of binary number are grouped into 4 bits starting from LSB.

Step 3: Combine all groups.

EXAMPLE:

$$\text{Convert}\left(1076\right)_8 = \left(?\right)_{16}$$

$$\left(1076\right)_8 = \quad 1 \quad 0 \quad 7 \quad 6$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

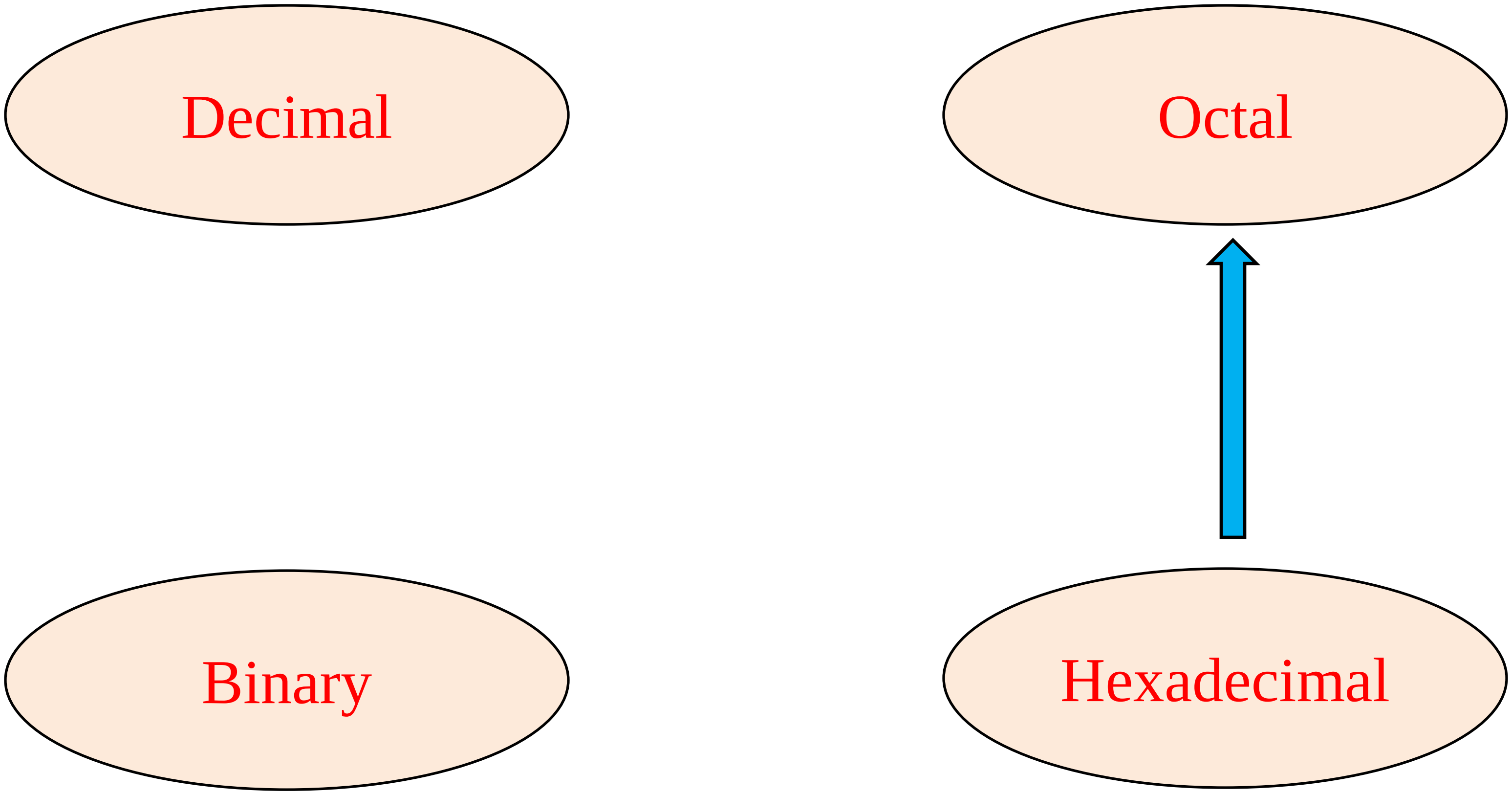
$$\underbrace{001\ 0}_{2}\underbrace{00\ 11}_{3}\underbrace{1\ 110}_{14}$$

$$2 \qquad 3 \qquad E$$

$$\left(1076\right)_8 = \left(23E\right)_{16}$$

CONVERSION OF NUMBER SYSTEM

HEXADECIMAL TO OCTALCONVERSION:



CONVERSION OF NUMBER SYSTEM

Step 1: Convert given hexadecimal number into binary number

Step 2: The bits of binary number are grouped into 3 bits starting from LSB.

Step 3: Combine all groups.

EXAMPLE:

1	D	0	E
0 0 0 1	1 1 0 1	0 0 0 0	1 1 1 0

0	0 0 1	1 1 0	1 0 0	0 0 1	1 1 0
	1	6	4	1	6

$(1D0E)_{16} = (16416)_8$

CONVERSION OF NUMBER SYSTEM

EXERCISE:

DECIMAL	BINARY	OCTAL	HEXADECIMAL
53			
	11001		
		143	
			2F3

CONVERSION OF NUMBER SYSTEM

EXERCISE:

DECIMAL	BINARY	OCTAL	HEXADECIMAL
53	110101	65	35
	11001		
		143	
			2F3

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EXERCISE:

DECIMAL	BINARY	OCTAL	HEXADECIMAL
53			
25	11001	31	19
		143	
			2F3

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EXERCISE:

DECIMAL	BINARY	OCTAL	HEXADECIMAL
53			
	11001		
99	1100011	143	319
			2F3

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EXERCISE:

DECIMAL	BINARY	OCTAL	HEXADECIMAL
53			
	11001		
		143	
755	1011110011	1363	2F3

CONVERSION OF NUMBER SYSTEM

EXERCISE:

DECIMAL	BINARY	OCTAL	HEXADECIMAL
53	110101	65	35
25	11001	31	19
99	1100011	143	319
755	1011110011	1363	2F3

THANK YOU

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